Project Proposal

Showroom Management System

Course: Data Structures and Algorithms (DSA)

Group Members

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Project Objectives

The objectives of this project are as follows:

To design a **Showroom Management System** that:

- Allows customers to view available cars and their details.
- Facilitates the purchase of cars from the showroom inventory.
- Tracks purchased cars and displays a customer's purchase history.
- Updates the car inventory and adds/removes cars.

To implement Custom Data Structures to manage the showroom's car inventory, customer data,and purchase history efficiently.

The system will utilize a combination of the following data structures:

- Array/List for storing car details.
- Hash Map for fast access to car details using the CarID.
- Binary Search Tree (BST) for sorting and searching cars based on attributes like
- price or model.
- Linked List for tracking customer purchases.

Technologies and Tools

- Programming Language: Java
- Data Structures: Custom implementations of arrays, hash maps, binary search trees, linked lists, and queues. These will be used to store and manipulate the car inventory, customer data, and purchase history efficiently.
- Exception Handling: Java's built-in Exception Handling (try, catch, throw) will be utilized to handle errors and ensure the program remains robust, even when faced with invalid inputs or unexpected conditions.
- User Interface: The system will feature a basic text-based interface (CLI) for interacting with users. For simplicity and efficiency, the interface will be menu-driven, allowing the user to easily access functionalities like viewing available cars, making purchases, and checking purchase history.

Key Features

The Showroom Management System will provide the following key features:

View Available Cars

- Functionality: Users can view a list of all cars available in the showroom, displaying details like CarID, Model, Brand, Price, Year, and Availability.
- Data Structure Used: Array/List will store all available cars. Sorting and filtering will be handled by the Binary Search Tree (BST) for attributes such a Price, Brand, and Year.

Search Cars by Attributes

- Functionality: Users can search for cars by different attributes, including Price Range, Brand, Model, or Year.
- Data Structure Used: Binary Search Tree (BST) will enable efficient searching and sorting of cars based on the selected attribute. The Hash Map will be used to quickly access and display detailed information of specific cars.

Purchase a Car

- Functionality: Customers can purchase a car from the showroom, after which the car will be removed from the available inventory.
- Data Structure Used: Queue will manage the purchase requests in FIFO (First In, First Out) order. The Linked List will store each customer's purchase history, including car details and the purchase date.

View Purchased Cars

- Functionality: Customers can view a list of cars they have purchased, along with the purchase details.
- Data Structure Used: Linked List will store the purchase history for each customer, where each node represents a car purchased by the customer. View Detailed Information of a Car
- Functionality: By selecting a car from the available inventory, customers can view detailed information like Price, Model, Brand, Year, and Availability.
- Data Structure Used: Hash Map will be used to quickly access detailed car information using the CarID as the key.

Manage Car Inventory

- Functionality: Admin users can add new cars to the showroom inventory or remove cars that have been sold.
- Data Structure Used: Array/List will be used to maintain the inventory list. Updates to the inventory will be reflected by adding or removing cars from the list.

Sort Cars by Price or Year

- Functionality: Users can sort cars based on different criteria like Price or Year.
- Data Structure Used: Binary Search Tree (BST) will be used to keep the cars sorted automatically. Alternatively, Array/List could be sorted when needed.

Handle Exception Scenarios

- Functionality: The system will have built-in exception handling for scenarios like invalid user inputs, unavailable cars, or unsuccessful purchases.
- Technology Used: Java's Exception Handling (try, catch, throw) will ensure that the system remains robust and continues to function smoothly even when unexpected errors occur (e.g., trying to purchase a car that is already sold).

Customer Profile and Purchase History

- Functionality: Each customer will have a profile to track their purchases. They can view their purchase history and any other personal details (e.g., name, contact details).
- Data Structure Used: Linked List will store each customer's purchase history.

Conclusion

The **Showroom Management System** will provide a hands-on learning experience in the application of custom data structures for managing real-world systems. By integrating arrays, hash maps, binary search trees, linked lists, and queues, the project will showcase the ability to handle car inventory, customer purchases, and other related operations in a well-structured and efficient manner. This system will serve as a robust solution for a showroom environment, demonstrating the use of fundamental data structures and algorithms to build practical applications.